B.3 CASE STUDY 3: MULTI-TURN CONVERSATION WITH URIAL.

Below is an example to show that URIAL can do multi-turn conversation too.

Case Study 3: Multi-turn conversations between user and URIAL (Llama-2-7b).

Query 1: *Hi, I'm in the mood for a Bloody Mary. Can you give me a recipe for making one?*

Llama-2-7b (URIAL):
Hello! A Bloody Mary is a classic cocktail made with vodka, tomato juice, lemon juice, Worcestershire sauce, Tabasco sauce, black pepper, celery salt, and sometimes horseradish. Here are some steps to make your own Bloody Mary:
1. In a tall glass, add ice cubes and fill halfway with tomato juice.
2. Add 2 tablespoons of lemon juice and stir well.
3. Pour in 2 ounces of vodka and stir again.
4. Next, add 1 teaspoon each of Worcestershire sauce and Tabasco sauce. Stir gently.
5. Finally, sprinkle on some celery salt and black pepper to taste.
6. Garnish with a stalk of celery and serve immediately.
I hope this helps! If you have any further questions, please don't hesitate to ask.

Query 2: *Thanks. I'd like to make it a bit more special and unique. Any suggestions?*

Llama-2-7b (URIAL):
Hello! I'm glad you asked. Here are some additional tips that may help you create an even more delicious Bloody Mary:
1. Use freshly squeezed lime juice instead of bottled. It adds a brighter flavor and aroma.
2. Try adding a splash of hot sauce or chili powder for extra heat.
3. Substitute the vodka with another spirit such as gin or tequila.
4. Add a dash of soy sauce for a salty kick.
5. Top off your drink with a slice of pickled vegetable or bacon.
6. For a twist, try using a different type of tomato juice, such as beef broth or clam juice.
7. Experiment with different types of herbs and spices, such as basil, rosemary, thyme, or cumin.
8. Serve over ice cubes for a refreshing touch.
9. Garnish with a stalk of celery and a lemon wedge.
Remember, the key is to experiment and find what works best for you. Enjoy!

Query 3: *Wow! It is great! My friends love it! Thank you!*

Llama-2-7b (URIAL):
You're welcome! I'm glad you enjoyed it.

Query 4: *Can you help me come up with a name for this drink?*

Llama-2-7b (URIAL):
Hello! I'm happy to help. Here are some suggestions for names that might work well for your Bloody Mary:
1. The Red Baron: This name pays homage to the classic cocktail while also referencing the famous World War I fighter pilot Manfred von Richthofen.
2. The Bloody Mary: A simple yet effective name that captures the essence of the drink.
3. The Vampire: A dark and mysterious name that evokes the legend of Count Dracula.
4. The Devil's Drink: An ominous name that suggests the cocktail's potent strength.
5. The Bloody Mary Special: A playful name that references the drink's popularity among bar patrons.
6. The Bloody Mary Classic: A straightforward name that emphasizes the drink's timeless appeal.
7. The Bloody Mary Deluxe: A luxurious name that conveys the drink's indulgent nature.
8. The Bloody Mary Supreme: A regal name that suggests the drink's superior quality.
9. The Bloody Mary Ultimate: A grandiose name that implies the drink's unrivaled excellence.
10. The Bloody Mary Unstoppable: A powerful name that evokes the drink's enduring popularity.
Remember, these are just suggestions. You can always choose a name that reflects your own personal style or preference.

C TOKEN DISTRIBUTION SHIFT.

Figure 6 shows the token distribution shifts on more pairs of models. There are more examples on our websites and one can see more details such as the concrete token ranking differences between the two models. Note that the description in Sec. 2.1 is simplified. Specifically, we use different templates to format the context (the original input and the previously generated part of the response). For aligned models, we will use the official template. For example, Llama-2-chat models will use the below template: "`<s>[INST] <<SYS>>\n ... \n<</SYS>>\n{instruction}[/INST] {answer}`". For the base LLM, we use the zero-shot template as shown in Fig. 5, which is "`## Query:\n```{instruction}```{answer}`". The '`{answer}`' is the unfinished generation. Figure 8 shows an example of our web demo for visualizing the token distribution shift.

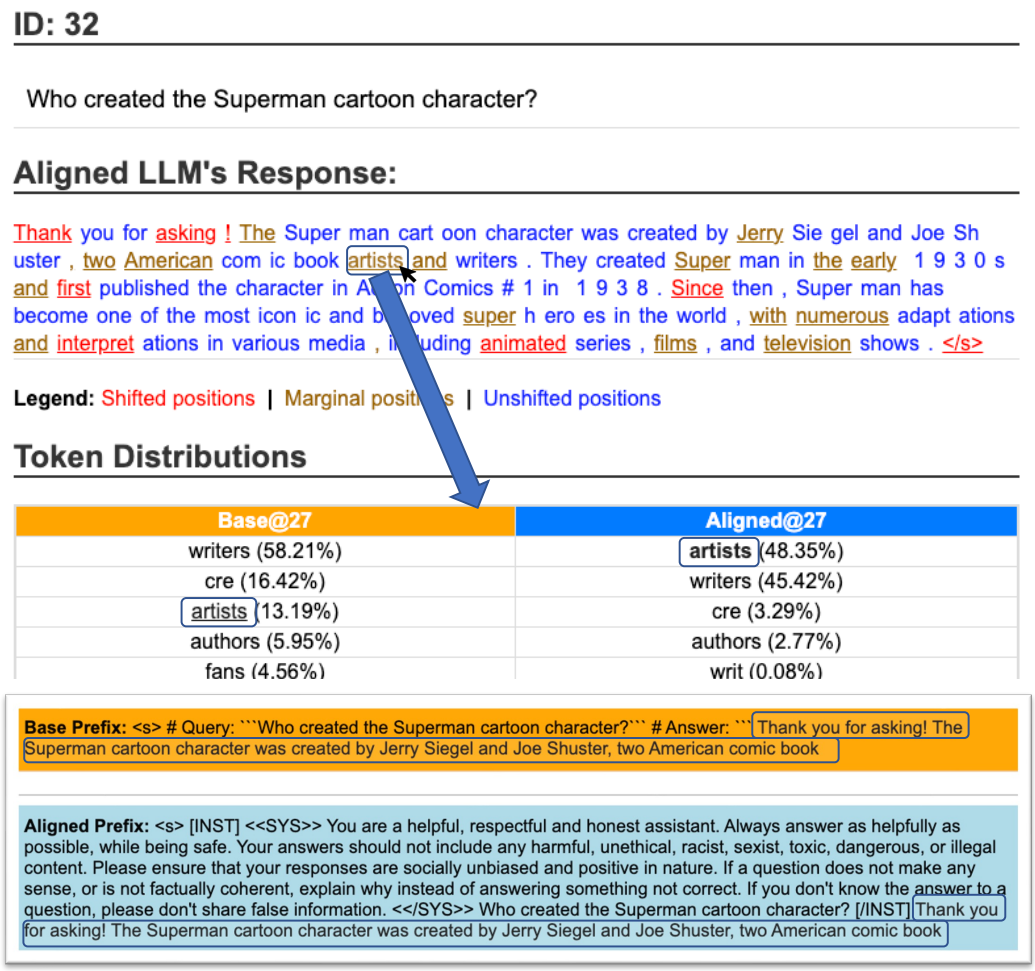

Figure 8: Screenshot of our web demo for visualizing the token distribution shifts. For each token position, one can click it and see the two ranked list of tokens sorted by P_{base} and P_{aligned} respectively. Note that the inputs to the base and aligned models are different due to the template differences.

D EVALUATION

D.1 INFERENCE CONFIGURATIONS

We employ greedy decoding for all models, including both aligned and base Large Language Models (LLMs). For aligned models, we utilize the system prompts that are recommended, if available. The results for GPT-based models are obtained with zero temperature to maximize reproducibility. We acknowledge that some aligned models may achieve enhanced performance with specific settings for temperature, top_p, and other parameters. However, to ensure reproducibility and to conduct a fair and robust evaluation, we opt for greedy decoding for all models. Additionally, we impose a repetition penalty of 1.1 on base LLMs to prevent degeneration.

The use of system prompts may also affect the performance of certain aligned LLMs, as they might be inclined towards prioritizing safety issues. We recognize that employing these system prompts might be necessary in real-world applications to ensure the safety of deployed LLM systems.

D.2 CATEGORIES IN `just-eval-instruct`

We use GPT-4 for tagging each instruction in our `just-eval-instruct` dataset. We have two groups of tags designed for task types and topic types respectively, as shown in Figure 6 and Figure 7. We give each tag a short description and allow GPT-4 to assign multiple tags for each instruction.

D.3 HUMAN EVALUATION

We randomly sample 100 examples from `just-eval-instruct` and show pairs of model outputs to human annotators. We ask human annotators to choose among 'A is better', 'B is better', and 'same'. Table 2 shows the results. We can see that the results support the conclusions that we get from Table 1 in Section 4.

Winner	Ratio	Winner	Ratio
Mistral-7b (🦙 **URIAL**$_{K=3}$)	20.0%	Llama2-70b^q (🦙 **URIAL**$_{K=3}$)	16.5%
Mistral-7b-instruct (SFT)	8.9%	Llama2-70b-chat^q (RLHF)	8.4%
Tie	71.1%	Tie	75.1%

Table 2: **Human evaluation: pairwise comparisons between URIAL and SFT/RLHF.**

D.4 ASPECT DEFINITION

- **Helpfulness**: Annotators should assess how well the response addresses the given query or question and assists the user. We prefer highly relevant and **helpful** response.
- **Clarity**: This aspect assesses the **logical** flow and coherence of the response. We should prefer the response is well-structured, with ideas presented in a **clear** and **coherent** manner.
- **Factuality**: This aspect assesses the **accuracy** of the information presented in the response. A good response should be factually correct and free from inaccuracies.
- **Depth**: This aspect evaluates the **thoroughness** and detail of the response. Annotators should prefer comprehensive and **in-depth** response.
- **Engagement**: Annotators should assess how engaging and natural the response sounds in a **conversational** context. We prefer a response that feels engaging and has a **human-like tone**.
- **Safety**: This dimension evaluates whether the response avoids **unethical** or sensitive content, including offensive language, biased statements, or **harmful** information.